Appendix –A (Dynamics of the robot)

Leg 1 as stance leg

The distance of the center of gravity of leg 1 in the horizontal and vertical directions from the pivot point is given as follows (Figure 1b of the manuscript)

$$x_1 = \frac{L}{2}\cos\left(\frac{\pi}{2} - \theta_1\right) \tag{A.1}$$

$$y_1 = \frac{L}{2}\sin\left(\frac{\pi}{2} - \theta_1\right) \tag{A.2}$$

Similarly, the distance of the center of gravity of leg 2 in the horizontal and vertical direction is given as follows

$$x_2 = L\sin(\theta_1) - \frac{L}{2}\cos(\theta_2) \tag{A.3}$$

$$y_2 = L\cos(\theta_1) - \frac{L}{2}\cos(\theta_2) \tag{A.4}$$

The distance of the center of gravity of the hip in the horizontal and vertical direction is given as follows

$$x_H = L\cos\left(\frac{\pi}{2} - \theta_1\right) \tag{A.5}$$

$$y_H = L\sin\left(\frac{\pi}{2} - \theta_1\right) \tag{A.6}$$

Total Kinetic Energy is given as

$$\begin{aligned} T = {} & \frac{1}{2}m\left(\left[\frac{d}{dt}(x_1)\right]^2 + \left[\frac{d}{dt}(y_1)\right]^2\right) + \frac{1}{2}m\left(\left[\frac{d}{dt}(x_2)\right]^2 + \left[\frac{d}{dt}(y_2)\right]^2\right) \\ & + \frac{1}{2}m_H\left(\left[\frac{d}{dt}(x_H)\right]^2 + \left[\frac{d}{dt}(y_H)\right]^2\right) \end{aligned} \tag{A.7}$$

$$U = m\ g\ y_1 + m\ g\ y_2 + m_H y_H \tag{A.8}$$

The Lagrange Γ is given as

$$\Gamma = T - U \tag{A.9}$$

Using Euler-Lagrange equations

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial \Gamma}{\partial \dot{\theta}_1}\right) - \frac{\partial \Gamma}{\partial \theta_1} &= \tau_1 \\ \frac{d}{dt}\left(\frac{\partial \Gamma}{\partial \dot{\theta}_2}\right) - \frac{\partial \Gamma}{\partial \theta_2} &= \tau_2 \end{aligned} \tag{A.10}$$

$$M_1(\theta_1,\theta_2)\begin{bmatrix}\ddot{\theta}_1 \\ \ddot{\theta}_2\end{bmatrix} + D_1\left(\theta_1,\theta_2,\dot{\theta}_1,\dot{\theta}_2\right)\begin{bmatrix}\dot{\theta}_1 \\ \dot{\theta}_2\end{bmatrix} + K_1(\theta_1,\theta_2) = \begin{bmatrix}\tau_1 \\ \tau_2\end{bmatrix} \tag{A.11}$$

$$M_1(\theta_1,\theta_2) = \begin{bmatrix} -L^2\left(\frac{5}{4}m + m_H\right) & \frac{1}{2}mL^2\cos(\theta_1-\theta_2) \\ \frac{1}{2}mL^2\cos(\theta_1-\theta_2) & \frac{-1}{4}mL^2 \end{bmatrix} \tag{A.12}$$

$$D_1(\theta_1,\theta_2) = \begin{bmatrix} 0 & \frac{1}{2}mL^2\sin(\theta_1-\theta_2)\dot{\theta}_2 \\ \frac{-1}{2}mL^2\sin(\theta_1-\theta_2)\dot{\theta}_1 & 0 \end{bmatrix} \tag{A.13}$$

$$K_1(\theta_1,\theta_2) = \begin{bmatrix} gL\left(\frac{3}{2}m + m_H\right)\sin(\theta_1) \\ \frac{-1}{2}gLm\sin(\theta_2) \end{bmatrix} \tag{A.14}$$

Leg 2 as stance leg

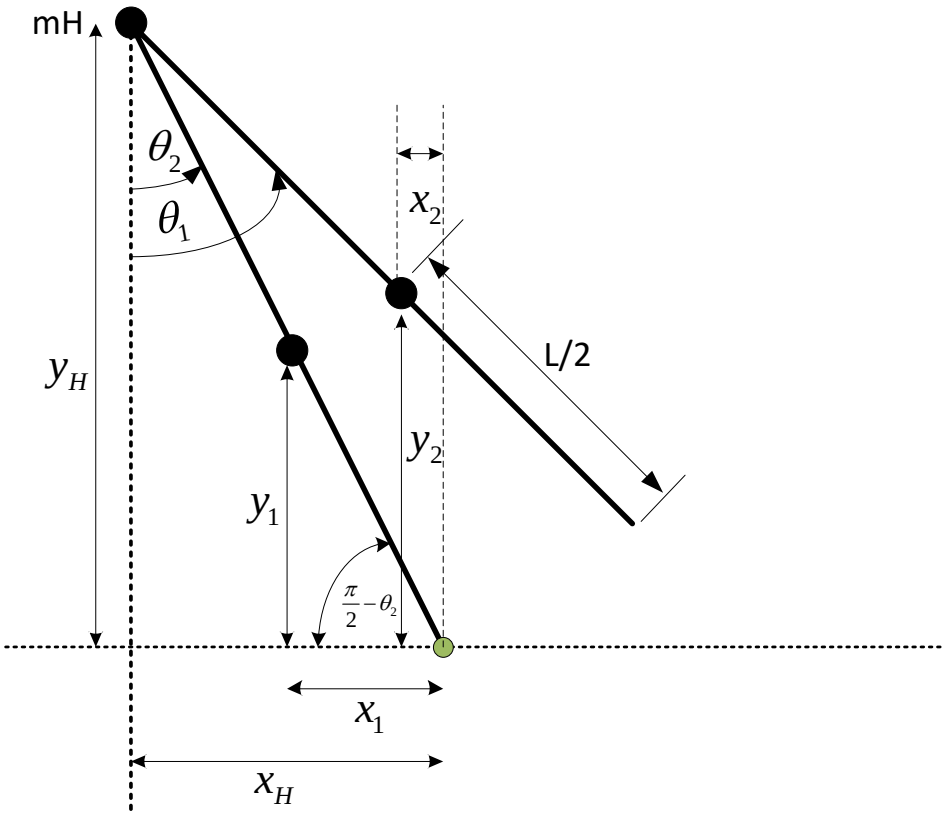


Figure A.1 Nomenclature of the bipedal robot

The distance of the center of gravity of leg 1 in the horizontal and vertical direction from the pivot point is given as follows (Figure A.1)

$$x_1 = \frac{L}{2}\cos\left(\frac{\pi}{2}-\theta_2\right) \tag{A.15}$$

$$y_1 = \frac{L}{2}\sin\left(\frac{\pi}{2}-\theta_2\right) \tag{A.16}$$

Similarly, the distance of the center of gravity of leg 2 in the horizontal and vertical direction is given as follows

$$x_2 = L\sin(\theta_2)-\frac{L}{2}\cos(\theta_1) \tag{A.17}$$

$$y_2 = L\cos(\theta_2) - \frac{L}{2}\cos(\theta_1) \tag{A.18}$$

The distance of the center of gravity of the hip in the horizontal and vertical direction is given as follows

$$x_H = L\cos\left(\frac{\pi}{2} - \theta_2\right) \tag{A.19}$$

$$y_H = L\sin\left(\frac{\pi}{2} - \theta_2\right) \tag{A.20}$$

The Lagrange Γ is given as

$$\Gamma = T - U \tag{A.21}$$

Using Euler Lagrange equations

$$\begin{aligned} \frac{d}{dt}\left(\frac{\partial\Gamma}{\partial\dot{\theta}_1}\right) - \frac{\partial\Gamma}{\partial\theta_1} &= \tau_1 \\ \frac{d}{dt}\left(\frac{\partial\Gamma}{\partial\dot{\theta}_2}\right) - \frac{\partial\Gamma}{\partial\theta_2} &= \tau_2 \end{aligned} \tag{A.22}$$

$$M_2(\theta_1,\theta_2)\begin{bmatrix}\ddot{\theta}_1\\ \ddot{\theta}_2\end{bmatrix} + D_2(\theta_1,\theta_2,\dot{\theta}_1,\dot{\theta}_2)\begin{bmatrix}\dot{\theta}_1\\ \dot{\theta}_2\end{bmatrix} + K_2(\theta_1,\theta_2) = \begin{bmatrix}\tau_1\\ \tau_2\end{bmatrix} \tag{A.23}$$

$$M_2(\theta_1,\theta_2) = \begin{bmatrix} \frac{-1}{4}mL^2 & \frac{1}{2}mL^2\cos(\theta_1-\theta_2) \\ \frac{1}{2}mL^2\cos(\theta_1-\theta_2) & -L^2\left(\frac{5}{4}m + m_H\right) \end{bmatrix} \tag{A.24}$$

$$D_2(\theta_1,\theta_2) = \begin{bmatrix} 0 & \frac{1}{2}mL^2\sin(\theta_1-\theta_2)\dot{\theta}_2 \\ \frac{-1}{2}mL^2\sin(\theta_1-\theta_2)\dot{\theta}_1 & 0 \end{bmatrix} \tag{A.25}$$

$$K_2\left(\theta_1,\theta_2\right)=\begin{bmatrix} \frac{-1}{2}gLm\sin(\theta_1) \\ gL\left(\frac{3}{2}m+m_H\right)\sin(\theta_2) \end{bmatrix} \tag{A.26}$$